Status	Finished
Started	Tuesday, 20 October 2026, 11:19 AM
Completed	Tuesday, 20 October 2026, 11:24 AM
Duration	4 mins 16 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Linear

c. Hash

d. Binary ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search is more efficient than binary search on large lists.

c. Linear search divides the list into halves.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search

b. Hash search

c. Interpolation search

d. Linear search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. n-1

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By comparing each element in the list sequentially ✓
b. By sorting the list first
c. By using a hash function
d. By dividing the list into halves

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. Item is the last element in the array or item is not there at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is not the array at all

Question **7**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(1) ✓
c. O(log n)
d. O(n log n)

Question **8**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The list is sorted
c. The search continues in the left sublist
d. The search continues in the right sublist

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist
b. There must be a mechanism to delete and/or insert elements in the list ✓
c. Number values should only be present
Number values should only be present
d. The list must be sorted

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O(1)
d. O🦴 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the left sublist ✓
c. In the entire list
d. At the beginning of the list

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Dynamic search
c. Linear search ✓
d. Binary search

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O🥊
c. O(n log n)
d. O(1)

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Sorting the list before searching
d. Checking each element sequentially ✓

Question **15**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓
b. Insertion
c. Selection
d. Complexity